Code-Switching as a Controlled Variable: Hebrew–English Mixing Degrades Sentiment Classification and Displaces Representations

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Abstract

Israeli technology workplaces communicate in continuously mixed Hebrew and English, yet code-switching is normally treated in NLP as a binary corpus property: text either is code-mixed or it is not. We instead treat mixing as an *ordered, controlled variable*. We construct a Hebrew–English code-switched sentiment dataset in which 105 seed sentences are each rendered at four mixing levels (0/30/60/100% English) with propositional content held constant, so that differences across levels can be interpreted primarily as effects of the controlled mixing manipulation. Fine-tuning two encoders — the multilingual mmBERT and the Hebrew-specific DictaBERT — we find that models trained only on pure Hebrew degrade significantly as mixing rises (Cochran–Armitage $p = 0.042$ and $p = 0.019$ respectively), with the sharpest fall between 30% and 60% mixing, where DictaBERT's accuracy approaches the three-class chance level. Training on mixed data recovers most of the loss (pooled accuracy $0.59 \rightarrow 0.94$ and $0.42 \rightarrow 0.84$, Fisher $p < 10^{-5}$) and we find no statistically significant degradation trend under mixed training. In our experimental setting, the Hebrew-*specific* model is the more fragile of the two at every mixing level, and it also drifts roughly twice as far in embedding space; across models and mixing levels, embedding drift tracks accuracy loss ($\rho = 0.93$, $n = 6$). Hebrew specialisation thus appears to reduce robustness to code-switching in this setting, and the results are consistent with representational displacement as a contributing mechanism. A further, task-free check probes whether this is sentiment-specific: does the model still recognise two mixing-level variants as the same sentence? That signal degrades over the identical 30–60% window with no labelled data required, providing complementary evidence that the effect is not confined to the sentiment classification setup.

1 Problem and Motivation

Hebrew–English code-switching is common in Israeli technology workplaces. A single Slack message may carry Hebrew syntax with English technical vocabulary (*ha-deployment karas be'emtsa ha-laila* — "the deployment crashed in the middle of the night", where only *deployment* switches), English syntax with Hebrew adverbials, or an unpredictable alternation of both. NLP systems deployed on this text — sentiment monitoring, incident triage, employee feedback analysis — are typically evaluated either on clean Hebrew or on clean English, and their reported accuracy may therefore substantially overstate field performance.

Existing code-switching resources treat mixing as a fixed property of a corpus. This obscures the question a practitioner actually faces: *how much* mixing is tolerable before a model fails, and does the failure arrive gradually or as a cliff? Answering that requires varying mixing while holding meaning fixed — an experimental manipulation that naturally-occurring corpora cannot provide, because in real data the amount of mixing is confounded with topic, speaker, and register.

We therefore ask four questions:

1. Does sentiment accuracy degrade as Hebrew–English mixing increases, and if so, where is the degradation sharpest?
2. Is a Hebrew-specialised encoder more or less robust to code-switching than a general multilingual one?
3. Is accuracy loss explained by displacement of the sentence in the model's representation space?
4. Is any of this specific to sentiment, or does code-switching disrupt NLP tasks more generally?

2 Related Work

Code-mixed sentiment. The closest prior work is SemEval-2020 Task 9, SentiMix [5], which provides sentence-level sentiment over Hinglish (Hindi–English) and Spanglish (Spanish–English) code-mixed tweets, approximately 20K and 19K items respectively. The best submitted systems reached 75.0 F1 on Hinglish and 80.6 on Spanglish. SentiMix establishes that code-mixed sentiment is harder than monolingual sentiment, but treats code-mixing as a binary attribute of each item. Our contribution relative to SentiMix is the *graded* framing: by holding content constant and varying only mixing proportion, we convert "code-mixed text is harder" into a dose–response curve and can localise *where* in that range the failure occurs.

Code-switching benchmarks. GLUECoS [3] is the standard evaluation suite for code-switched NLP, spanning language identification, POS, NER, sentiment, QA, and NLI for English–Hindi and English–Spanish. Its central finding — that multilingual models fine-tuned on *artificially generated* code-switched data outperform all alternatives — is one we independently reproduce on a new language pair. LinCE [1] offers a comparable centralised benchmark. Neither covers Hebrew–English.

Models. mmBERT [4] is a ModernBERT-based multilingual encoder pretrained on 3T tokens spanning over 1800 languages, and is the first multilingual encoder to improve on XLM-R. For Hebrew, DictaBERT [6] and AlephBERTGimmel [2] are strong Hebrew pretrained encoders. To our knowledge, neither Hebrew model has previously been evaluated on Hebrew–English code-switched input.

3 Dataset

Construction. We wrote 105 seed sentences in Israeli tech workplace registers — deployments, standups, code review, incidents, sprints, hiring, infrastructure — balanced across positive, negative, and neutral sentiment. Each seed was rendered at four mixing levels:

Level	Description
0%	Pure Hebrew
30%	Hebrew matrix, English technical nouns
60%	Frequent alternation between Hebrew and English
100%	Pure English

yielding $105 \times 4 = 420$ items. Because the four variants of a seed express the same proposition, mixing level functions as a manipulated independent variable.

Generation. The variants are *LLM-generated renderings* of hand-written seeds, not scraped Slack logs. This is a deliberate trade: it buys clean experimental control over mixing level at the cost of natural distributional realism.

Annotation. Every item's sentiment label was collected via two independent passes, both blind to mixing level and to each other's answers: a human reviewer (the author) and an LLM reviewer (Claude, Anthropic), with the human reviewing and adjudicating disagreements before the final label was fixed. Agreement statistics between the two reviewers are reported in Section 5.6.

Splits. We split *by seed*, not by row: all four mixing-level variants of a seed are assigned to the same split (73/16/16 seeds → 292/64/64 items), stratified by sentiment, and verified to have zero seed overlap between splits — since the four variants of a seed are near-paraphrases of each other, this ensures no near-paraphrase of a test item appears in training.

4 Method

Embedding geometry. For each seed and each of six encoders, we extract embeddings (mean-pooled final hidden states for masked-LM encoders, native `encode()` for sentence-transformers) and compute the cosine distance between the 0% variant and each mixed variant. Since meaning is constant within a seed, this distance measures representational displacement associated with the controlled mixing manipulation.

Fine-tuning. We fine-tune mmBERT and DictaBERT for 3-way sentiment classification under two conditions:

- **Condition A** — trained on 0% (pure Hebrew) items only ($n = 73$).
- **Condition B** — trained on the full mixed training set ($n = 292$).

Both are evaluated on the identical held-out test set, broken out by mixing level. Hyperparameters are held fixed across all runs for comparability (5 epochs, batch 16, lr 2×10^{-5}, weight decay 0.01, warmup ratio 0.1, max length 64, seed 42), selecting the best checkpoint by validation macro-F1.

Statistics. Per-mixing-level test cells contain only $n = 16$ items, so we report 95% Wilson score intervals rather than bare point estimates and avoid interpreting individual cells. The two claims that carry the paper are tested on pooled data: condition A versus B via Fisher's exact test ($n = 64$ per condition), and degradation across mixing level via the Cochran–Armitage trend test.

Generality check (Q4). A cheap-to-add, label-free test probes whether the above is sentiment-specific. *Semantic equivalence* requires no new labels or training: since a seed's four variants are paraphrases by construction, we treat a seed's own $(0\%, L\%)$ pair as a positive (same-meaning) example and a random *different* seed's $(0\%, L\%)$ pair as a negative example, use cosine similarity from the Phase 2 embeddings as a same/different score, and report ROC-AUC (0.5=chance, 1.0=perfect separation) at each mixing level L.

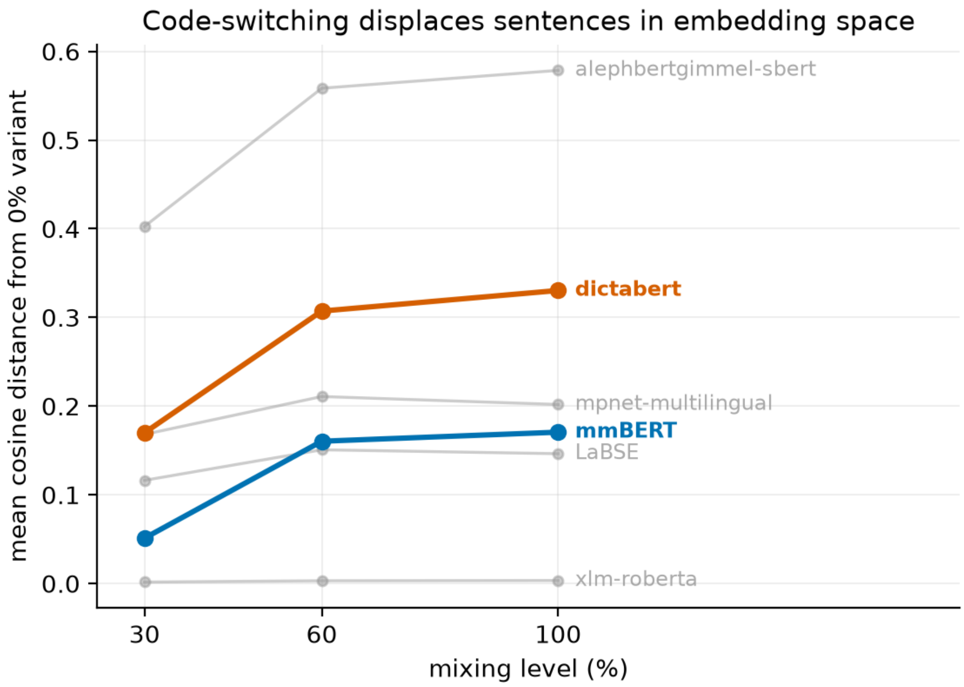

Figure 1: Mean cosine distance from each seed's 0% variant. All encoders drift monotonically with mixing level. The Hebrew-specific DictaBERT drifts roughly twice as far as the multilingual mmBERT.

Model	Cond.	Pooled acc.	95% CI
mmBERT	A	0.594	[0.47, 0.71]
mmBERT	B	**0.938**	[0.85, 0.98]
DictaBERT	A	0.422	[0.31, 0.54]
DictaBERT	B	0.844	[0.74, 0.91]

Table 1: Accuracy pooled across mixing levels ($n = 64$ per condition). Condition B beats A for both models: Fisher $p = 5.4 \times 10^{-6}$ (mmBERT), $p = 1.1 \times 10^{-6}$ (DictaBERT).

5 Results

5.1 Code-switching displaces representations

Figure 1 shows that every encoder displaces sentences monotonically as mixing increases. DictaBERT drifts about twice as far as mmBERT (0.330 versus 0.171 at 100%). XLM-R's near-zero distances (≈ 0.003) should *not* be read as robustness: mean-pooled masked-LM outputs are strongly anisotropic, compressing all cosine distances. Its t-SNE projection separates mixing levels cleanly despite the tiny absolute distances, so the geometry does encode mixing — raw cosine simply cannot resolve it for that model. Standard post-processing (mean-centering, or whitening the embedding covariance) is known to reduce anisotropy in masked-LM representations, but we did not apply it here; we therefore avoid directly comparing XLM-R's absolute cosine distances with those of the other encoders.

5.2 Pure-Hebrew training collapses under mixing

Under condition A, both models degrade significantly as mixing rises (Cochran–Armitage: mmBERT $z = -2.03$, $p = 0.042$; DictaBERT $z = -2.34$, $p = 0.019$). The steepest fall is between 30% and 60% mixing — mmBERT $0.69 \to 0.50$, DictaBERT $0.50 \to 0.31$ — indicating that

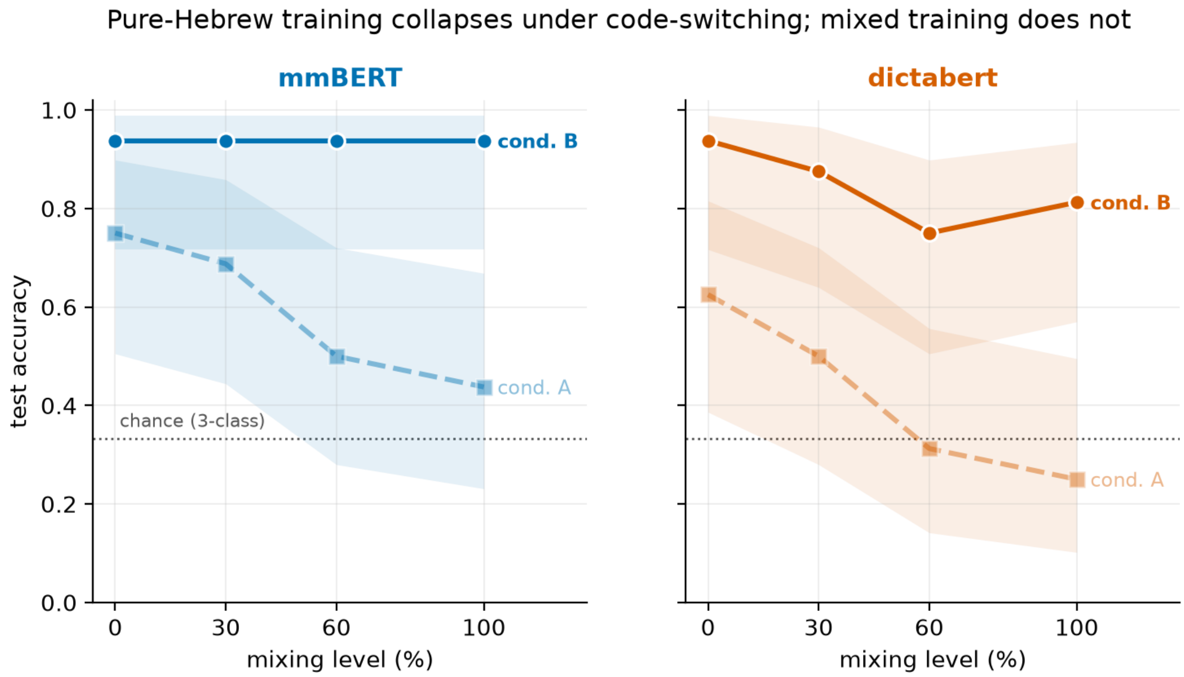

Figure 2: Test accuracy by mixing level. Shaded bands are 95% Wilson intervals ($n = 16$ per cell). Dashed = condition A (pure-Hebrew training), solid = condition B (mixed training). Dotted line marks 3-class chance.

the transition from primarily lexical borrowing to more frequent language alternation appears particularly challenging. At 60% and 100% mixing, DictaBERT's point estimates are close to the three-class chance level.

Condition B recovers most of the loss (Table 1), and we find no statistically significant degradation across mixing level under mixed training (mmBERT $p = 1.00$, DictaBERT $p = 0.24$). mmBERT becomes completely flat at 0.938 across all four levels. DictaBERT's apparent dip at 60% is *not* separable from noise (CI $[0.51, 0.90]$) and we do not interpret it.

5.3 Specialisation appears to be a liability in this setting

DictaBERT is worse than mmBERT at every mixing level in condition A, degrades on a steeper trend, and remains uneven even after mixed-data training. This is the opposite of the intuition that a Hebrew-tuned model should handle Hebrew workplace text best: the monolingual prior is precisely what English intrusion violates, whereas mmBERT's multilingual pretraining has already seen both sides of the switch. With only two models compared in one synthetic Hebrew–English setting, this should be read as evidence that specialisation reduced robustness *here*, not as a general claim about language specialisation.

5.4 Drift predicts accuracy loss

Pairing each model's embedding drift against its condition-A accuracy drop (Figure 3) gives $\rho = 0.93$. With six paired points this is descriptive rather than a powered test. It is, however, corroborated by an observation in the same direction: the model that drifts further (DictaBERT) is also the one that degrades further. Two complementary measurements showing the same qualitative pattern are consistent with representational displacement as a contributing mechanism behind the accuracy loss, though this does not establish a causal link.

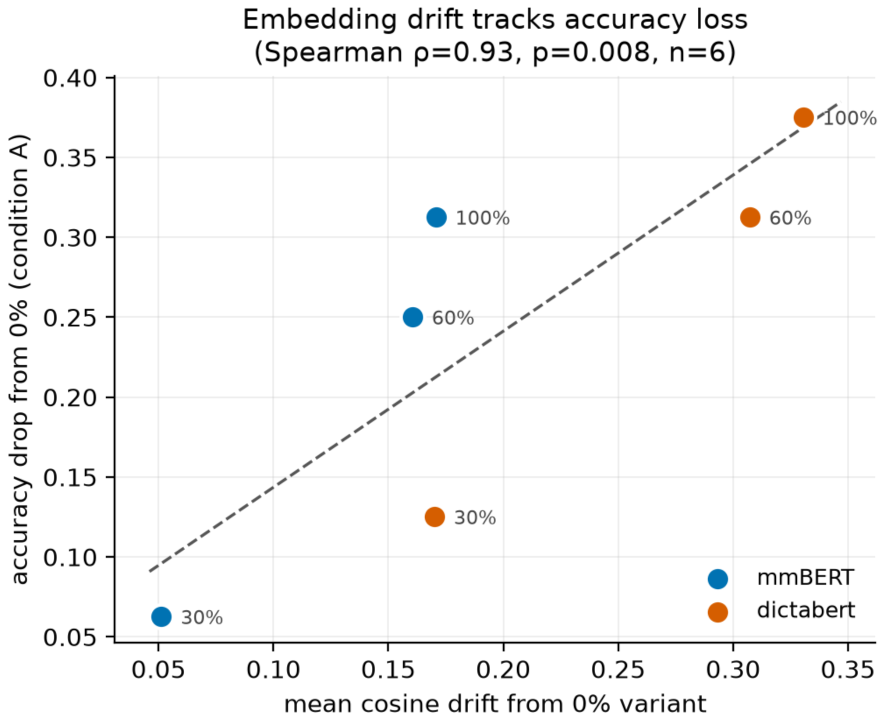

Figure 3: Embedding drift versus condition-A accuracy drop from each model's own 0% cell. Spearman $\rho = 0.93$ ($p = 0.008$, $n = 6$).

5.5 Is this specific to sentiment?

Semantic equivalence (Figure 4) needs no labels, only the embeddings already computed in Phase 2. All six encoders separate true paraphrases from unrelated sentences well above chance at every mixing level, but the two fine-tuned models decline over exactly the window identified above: mmBERT 0.95→0.78 and DictaBERT 0.98→0.89 from 30% to 60% mixing. Dedicated cross-lingual similarity models (mpnet, LaBSE) barely move, consistent with that being closer to their training objective. This provides complementary evidence, without depending on labeled data for a downstream task, that the observed representational changes are not confined to the sentiment classification setup — though this check still relies on the same underlying embeddings as §5.1, so it is not a fully independent replication.

5.6 Annotator ceiling

Both annotators (human and LLM) score 100% at every mixing level on the held-out test split. This is a genuine measurement but a weak ceiling: only 2 of 105 seeds drew any annotator error corpus-wide, and both fell in the training split. It indicates that our corpus encodes sentiment unusually unambiguously — a property of the dataset, not evidence that the task is easy in the wild.

6 Conclusion

Hebrew–English code-switching measurably degrades sentiment classification, and does so in a graded rather than binary fashion: accuracy is largely preserved under light lexical borrowing and falls sharply once alternation becomes structural, between 30% and 60% mixing. In our experimental setting, the Hebrew-specialised encoder is the more fragile of the two — in accuracy, in trend steepness, and in how far its representations move — suggesting that specialisation to

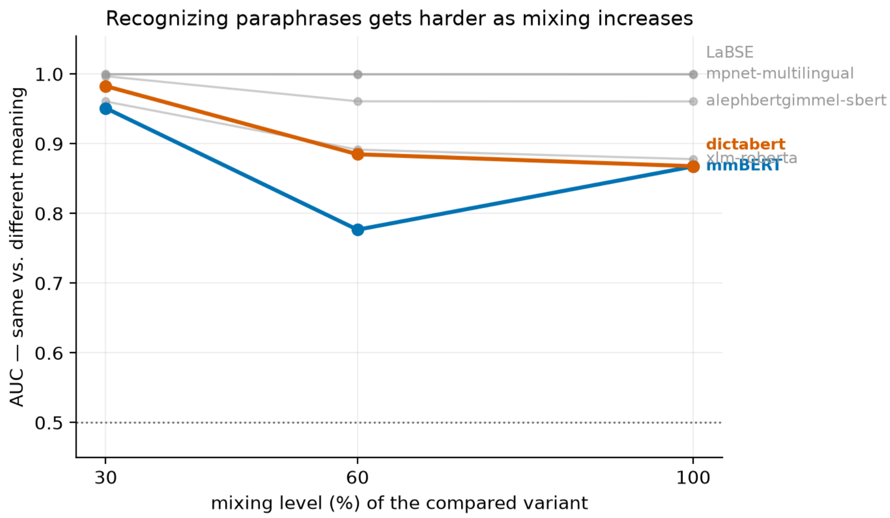

Figure 4: AUC for distinguishing a true paraphrase (same seed, different mixing level) from an unrelated sentence, using Phase 2 embeddings. No new labels or training.

one language can become a liability when the other intrudes. Fine-tuning on mixed data removes the measurable degradation, reproducing on Hebrew–English what GLUECoS reported for Hindi- and Spanish–English.

Extending beyond sentiment supports rather than narrows this account. A labeling-free semantic-equivalence check degrades over the identical 30–60% window with no labeled data involved, providing complementary evidence that representational drift, rather than an artifact of the sentiment classifier head, contributes to the effect.

The practical implication is direct: a Hebrew-tuned model validated on clean Hebrew will overstate its performance on real Israeli workplace text, and a simple and effective remedy in our setting is to place mixed-language data in the training set.

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